

ANALYTICS ENGINEERING / BI DEVELOPMENT / STAKEHOLDER REPORTING

One portfolio. Four versions of the truth.

Four teams. Four spreadsheets. One €977M portfolio. Every project meeting opened with 20 minutes of reconciling numbers before a single decision could be made. I built the system that ended that.

Power BI

DAX

Data Modelling

ETL Pipeline

Stakeholder Analytics

€977M Portfolio

708MW Capacity

4 Sources Unified

€977M

Total project budget tracked

708MW

Portfolio capacity monitored

4

Disconnected sources unified

0

Spreadsheets opened in meetings

Nobody was wrong. Nobody agreed.

The project controls team had a cost tracker. Finance owned the cashflow forecast. The project managers maintained their own progress files. Scheduling ran a separate resource plan. Every single one of those sources was accurate on its own terms.

The problem was that they could not talk to each other. The same project appeared under three different names across systems. Cost categories used different reference codes depending on who built the file. Monthly figures mixed confirmed spend with future projections sitting in the same column. There was no shared structure, so there was no shared truth.

WHAT I WALKED INTO

Every project meeting opened the same way. Teams would spend the first 20 minutes comparing figures from different files, explaining why the numbers looked different, and trying to agree on a starting point before anyone could actually discuss the project. Senior stakeholders were paying for that time every single week.

My job was to build a reporting system that collapsed all four pictures into one view. Not by forcing teams to change how they worked, but by handling the reconciliation in the data layer so it never had to happen in the meeting room again.

€977M

Total approved project budget

4

Separate sources with no shared structure

12

Active projects across 3 regions

708MW

Capacity under delivery

The real work happened before I opened Power BI.

Most people assume BI development starts with the visuals. It does not. Before I could build anything worth looking at, I had to make the data trustworthy. That meant going into the source files and fixing the structural problems that were causing every reconciliation headache in the first place.

01

Standardising project identities across sources

The same project had different names, codes and regional labels depending on which team's file you were in. I built a unified naming layer that resolved every project to a single consistent identity across all inputs. This was the foundation everything else depended on.

02

Separating actuals from forecasts

Monthly cost columns were combining confirmed historical spend with forward projections in the same field. I separated these using the reporting date as the boundary so the model always knew whether a value was a fact or an estimate. This distinction is what makes the cashflow visuals credible rather than misleading.

03

Reshaping wide format data into something usable

Month values arrived with one column per month, which works fine in a spreadsheet but breaks time-series calculations in a data model. I reshaped everything into long format so time intelligence functions, trend visuals and date-based filters could all work correctly without workarounds.

04

Writing DAX measures that work across all four domains

I wrote the calculation logic so the same measure could return the correct figure whether a user filtered by project, region, campus, date range or delivery stage. Budget variance, schedule performance, committed vs uncommitted spend and capacity delivery all came from one model without double-counting or breaking under filter combinations.

This is the part that does not show up in screenshots. But it is also the part that determines whether a reporting suite survives contact with real stakeholders or falls apart the first time someone applies an unexpected filter combination.

03 COMMERCIAL EXPOSURE

A €977M budget number tells you almost nothing useful on its own.

The commercial team's real question was never "what is the total." They already knew that. What they needed to see was how the money was actually distributed: how much was contractually locked in, how much was still uncommitted and therefore flexible, how much had converted to invoices, and whether contingency was being consumed at a healthy rate or quietly disappearing.

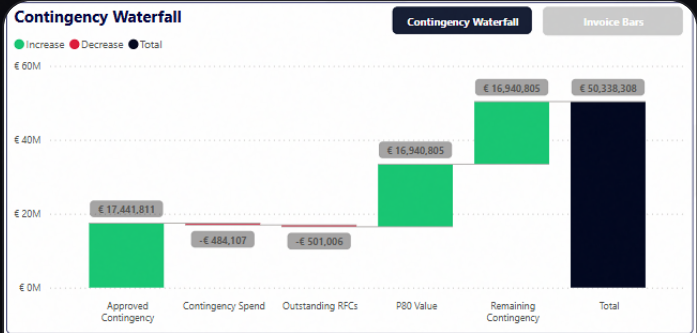


FIG 01 I used a waterfall to show contingency as a sequence of events rather than a final balance. Approved contingency, spend, outstanding change requests, risk-weighted value and remaining balance each get their own bar. The story of how that number was reached becomes readable at a glance.

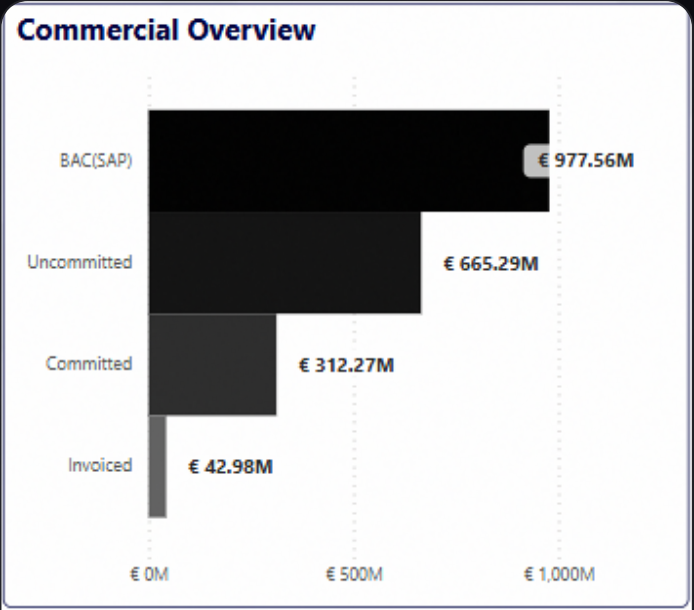


FIG 02 Total budget (€977M), committed spend (€312M), uncommitted exposure (€665M) and invoiced amount (€43M) in one view. The moment you see €43M invoiced against €977M total, you understand immediately where the cash flow risk sits.

I specifically chose the waterfall over a simple balance display because I noticed stakeholders were treating a healthy-looking remaining contingency figure as a green light, even when outstanding change requests were eating into it. Showing the movement rather than the endpoint changed the conversation. People stopped asking "are we okay" and started asking "which change requests are driving the reduction."

WHAT THE NUMBERS REVEAL

Only 4.4% of the total portfolio value had converted to invoiced spend at the point of this report. With €665M still uncommitted, the organisation had the majority of its purchasing decisions still ahead of it. Surfacing that gap explicitly gave the commercial team something concrete to manage, rather than a vague sense that spend would ramp up eventually.

Financial Summary							
DIN Summary	Category	BAC	Total Project Forecast	Committed	Uncommitted	Project Forecast EAC	Invoiced
100	Acquisition	€ 44,368,689.00	€ 44,368,680.00	€ 530,731	€ 43,837,958	€ 44,368,689.00	€ 384,597
200	Clearance and development	€ 66,982,509.00	€ 66,982,509.00	€ 19,001,820	€ 47,980,689	€ 66,982,509.00	€ 14,205,956
300	Structure - construction work	€ 246,829,578.00	€ 246,829,581.00	€ 82,276,529	€ 164,553,049	€ 246,829,578.00	€ 6,158,556
400	Building - Technical facilities	€ 532,978,206.00	€ 532,978,206.00	€ 176,870,633	€ 356,107,573	€ 532,978,206.00	€ 396,564
500	External Works	€ 6,453,318.00	€ 6,453,318.00	€ 2,151,105	€ 4,302,213	€ 6,453,318.00	€ 14,080
600	Furniture and Artwork	€ 0.00		€ 0	€ 0		€ 0
700	Ancillary Building costs	€ 46,472,211.00	€ 46,472,217.00	€ 14,701,879	€ 31,770,332	€ 46,472,211.00	€ 10,950,800
800	Client's responsibilities	€ 0.00		€ 0	€ 0		€ 0
Total		€ 977,563,107.00	€ 977,563,107.00	€ 312,271,995	€ 665,291,112	€ 977,563,107.00	€ 33,053,924

FIG 03 A category-level breakdown sitting behind the headline KPIs. I included this so analysts could verify any summary figure by tracing it back through cost categories without opening the source files. Dashboards earn trust when users can check the numbers, not just read them.

The design intent here was to make one report serve two audiences at the same time. Executives get the KPI cards and the bar chart. They have their answer in three seconds. Analysts get the detailed table. They have their audit trail without a separate file. One data model, two levels of depth, no extra spreadsheets needed.

I learnt early on that total spend figures are far less useful than timing.

The first time I sat in a finance review meeting, I watched a senior analyst treat a projected monthly figure as confirmed spend and start making funding assumptions based on it. The data was not wrong. The report just had no way of telling you which numbers were historical fact and which were forward estimates. Everything looked the same.



FIG 04 Cumulative cashflow from 2021 through 2026 with a vertical reporting date marker separating confirmed actuals from forward projections. The steep acceleration into 2024 and 2025 is now visible as a future funding pressure rather than buried in a row of monthly figures.

I added the reporting date marker specifically because of what I had watched happen without it. That single vertical line changes the entire reading of the curve. The question shifts from "how much has been spent" to "when is spend expected to accelerate and are we funded for that period." It prevented a category of mistake that had been happening in meetings regularly.

Cash Flow Matrix by DIN Code and Dates																	
Year	2021							2022									
DIN Code	August	September	October	November	December	Total	January	February	March	April	May	June	July	August	September	October	
100	0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 107,484	€ 67,606	€ 550	€ 1	
200	0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 293,618	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 261,1	
300	0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 154,1	
400	0	€ 0	€ 0	€ 0	€ 0	€ 2,500,000	€ 2,500,000	€ 0	€ 0	€ 0	€ 251,750	€ 0	€ 0	€ 0	€ 0	€ 609,1	
500	0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	
600	0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	
700	58	€ 350,450	€ 170,219	€ 740,915	€ 238,618	€ 2,247,722	€ 2,502,422	€ 379,488	€ 152,372	€ 328,527	€ 246,440	€ 63,847	€ 113,339	€ 174,998	€ 167,438	€ 647,829	€ 742,1
800	0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	
900	0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 69,1	
PC	0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	
Total	58	€ 350,450	€ 170,219	€ 740,915	€ 238,618	€ 2,747,722	€ 5,002,422	€ 673,107	€ 152,372	€ 328,527	€ 246,440	€ 315,597	€ 113,339	€ 282,482	€ 235,045	€ 648,379	€ 1,836,1

FIG 05 Monthly cashflow broken down by cost category code and year. I included this matrix as the verification layer behind the curve. Any analyst who questions a data point on the chart can trace it back to the exact month and category without leaving the report.

The matrix was a deliberate addition rather than an afterthought. I knew from experience that finance stakeholders are cautious about trusting a visual they cannot interrogate. Giving them the full monthly breakdown in the same report meant the curve was not just a picture. It was a verifiable statement backed by the underlying figures sitting directly beneath it.

05 PORTFOLIO CAPACITY

Leadership needed to see delivery risk, not a project list.

When I first looked at how portfolio capacity was being reported, it was essentially a table with project names and status labels. It told you the projects existed. It told you nothing about whether delivery was tracking against the baseline target or where the pipeline was heading across future financial years.

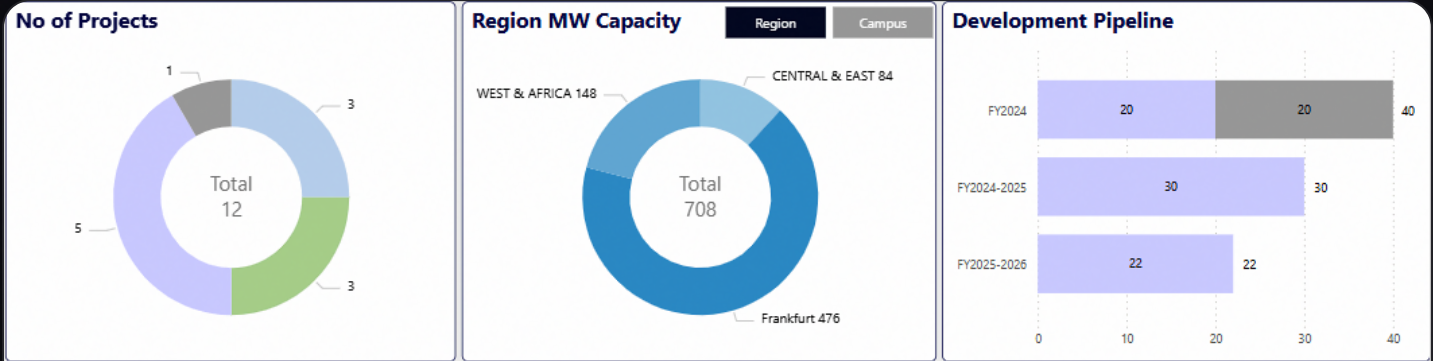
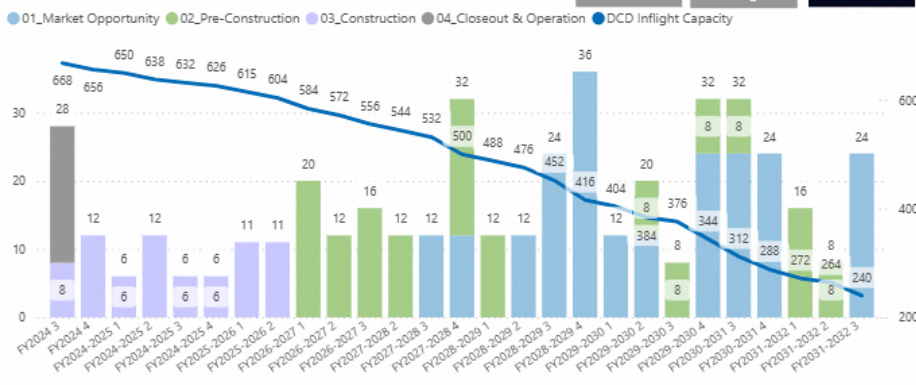


FIG 06 12 active projects delivering 708MW split across Frankfurt (476MW), West Africa (148MW) and Central East (84MW), alongside the development pipeline through FY2025-2026. The regional split was something I specifically surfaced because of what it implied about concentration risk.

When I mapped the regional breakdown, I noticed that Frankfurt was carrying 476MW out of 708MW total. That is 67% of all portfolio delivery sitting in one region. I flagged this to the portfolio team because it meant any delivery slippage in Frankfurt would not create a proportional impact on the overall numbers. It would create a disproportionate one. That was not visible anywhere in the previous reporting.

Capacity Delivery Per Quarter



RFO Lookahead

Campus	Building	Delivery Phase	IT Capacity (MW)	RFO Date (Baseline)	RFO Date (Forecast)	RAG
AMS1	C	1.1	11.00	26-Aug-25	26-Aug-25	G
AMS1	D	1.2	11.00	10-Jun-25	10-Jun-25	G
AMS1	E	1.1	12.00	30-Sep-27	30-Sep-27	G
AMS1	E	1.2	12.00	06-Jan-28	06-Jan-28	G
AMS1	E	1.3	12.00	27-Apr-28	27-Apr-28	G
BER2	C	12.00	12.00	18-Sep-28	18-Sep-28	G
BER2	C	12.00	12.00	12-Dec-28	12-Dec-28	G
BER2	C	12.00	12.00	20-Mar-29	20-Mar-29	G
BER3	A	12.00	12.00	27-Oct-27	27-Oct-27	G
BER3	A	12.00	12.00	02-Feb-28	02-Feb-28	G
BER3	A	12.00	12.00	17-Jan-29	17-Jan-29	G
BER3	A	12.00	12.00	11-Oct-29	11-Oct-29	G

FIG 07 Quarterly delivery curve with project stage breakdown and forward readiness lookahead table. I connected historical delivery performance to upcoming milestone dates in the same view so the conversation would always be about what was coming next rather than just what had already happened.

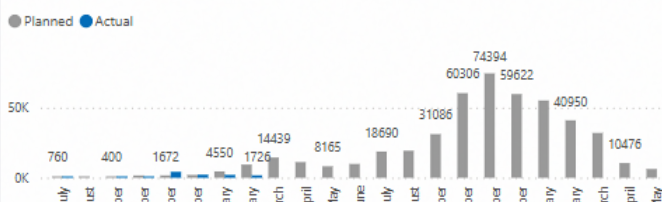
The lookahead table came from a direct request: stakeholders were tired of opening three separate files to understand where the portfolio stood. I consolidated the planned and forecast readiness dates alongside a risk indicator for each building into one view. A clean set of green indicators is reassuring. A single amber or red row changes the entire agenda of the meeting.

06 SCHEDULE PERFORMANCE

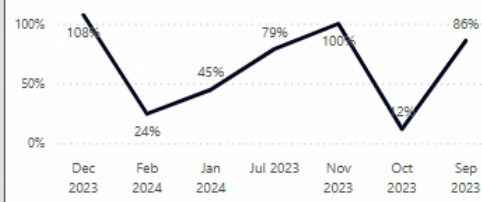
Hours being logged is not the same as work being on track.

Schedule reporting has a specific failure mode that I have seen in most project environments: it reports activity without reporting performance. Teams can be busy, hours can be accumulating and meetings can be running on time, while the project drifts further behind plan every single week. The numbers just do not surface the gap unless you build something that explicitly converts activity into a performance ratio.

Plan Vs Actual - Histogram



Schedule Performance Indicator (SPI)



% Tasks Complete

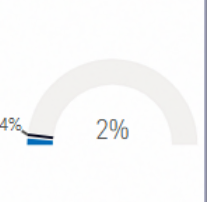


FIG 08 Plan versus actual histogram, Schedule Performance Index trend and task completion gauge in a single view. The SPI trend dropping to 24% in early 2024 before recovering to 86% is visible immediately. Without this reporting, that dip was invisible in the resource tracker.

The Schedule Performance Index converts raw hours into a ratio that tells you how much scheduled value is actually being delivered per unit of planned effort. An SPI of 62% means roughly 62 pence of progress for every pound of planned work. A variance of negative 7,017 hours means the gap is measurable, significant and growing.

I built the SPI trend line specifically because I wanted stakeholders to see movement over time rather than just a point-in-time figure. A single SPI reading tells you where you are. A trend line tells you whether the situation is improving, stabilising or deteriorating. Those are three very different conversations to be having in a project review.

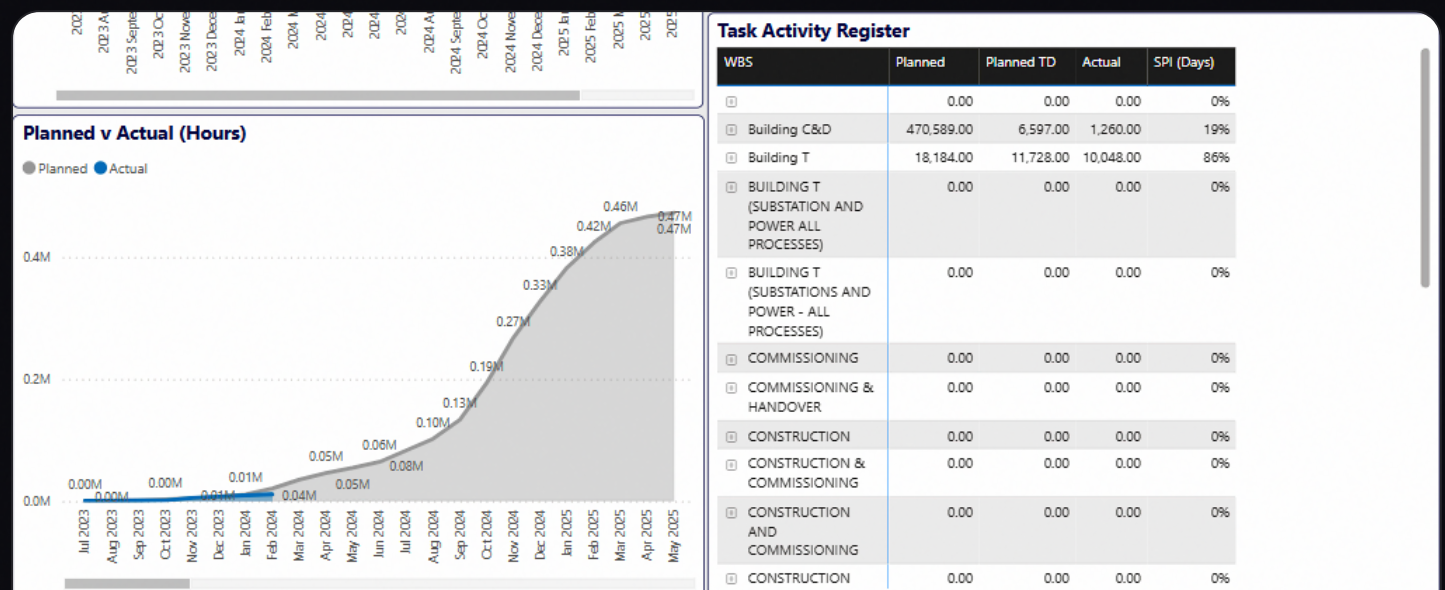


FIG 09 Planned versus actual hours trend and work package level activity register. Building C&D shows 470K hours planned against 1,260 actual hours logged at 19% schedule performance. That is the specific conversation the project manager needs to be having, now surfaced automatically rather than buried in a planner.

WHAT THIS CHANGED IN PRACTICE

Before this reporting existed, schedule reviews were retrospective summaries. Project managers would explain what had happened. After it, the reports identified the exception

before the meeting. The project manager walked in already knowing which work packages needed attention. The meeting became about solving the problem rather than finding it.

07 THE OUTCOME

Good reporting is measured by what stops happening.

The right way to evaluate any reporting system is not by counting charts or pages. It is by asking what problems it eliminated. Meetings stopped opening with reconciliation. Stakeholders arrived having already seen the current position. Decisions that previously required someone to pull three separate files and explain the difference between them could now be made by reading one screen.

The commercial team could track exactly where €977M in budget sat between committed, uncommitted and invoiced states at any point. Finance had a cashflow curve they could validate month by month without leaving the report. Portfolio leadership could see delivered capacity against the baseline and upcoming readiness milestones in a single view. Project managers could identify which work packages were driving schedule variance before anyone had to ask.

None of that required the teams to change how they worked. They kept their own files and their own processes. The reporting layer handled the reconciliation so the meetings could handle the decisions.

The skill underneath the construction context

The domain here is specific. The problem is not. Any organisation where decisions depend on data that lives across multiple teams in inconsistent formats has the same structural challenge. Building a trustworthy, unified reporting layer from fragmented inputs

